

MAT 6973 01T

Special Problems

Course Syllabus - Summer 2026



Course Information

Course Description: Prerequisite: Consent of instructor. An organized course offering the opportunity for specialized study not normally or not often available as part of the regular course offerings. Special Problems courses may be repeated for credit when topics vary, but not more than 6 hours, regardless of discipline, will apply to the Master's degree. This course has Differential Tuition. Course Fee: GS01 \$90.

Credit Hours: 3

Course Modality: Traditional in-person

Meeting Times

Duration: 05/26/2026-08/04/2026

Campus: Main Campus

Location: [MH 3.03.18](#)

Time(s): TR 11:00 AM - 12:45 PM

Learning Goals

At the end of the course, you will be able to:

- Construct and interpret algebraic invariants of topological spaces
- Produce rigorous proofs in algebraic topology
- Apply structural theorems to compute and classify
- Explain the correspondence between covering spaces and group theory
- Communicate mathematics independently on a self-directed topic

Communicate with Me

Instructor Name: Stephen Pena

Department

Mathematics

Office Location

FLN 4.01.22

Student Hours

T/R: 1:00-2:00

Email Address: stephen.pena@utsa.edu

Preferred Method of Communication

Office hours/speaking with me, followed by outlook email

About Me & My Teaching Philosophy

I believe mathematics is learned by doing, not by watching. My job is to guide you through difficult ideas and make sure you have the tools to wrestle with them yourself; your job is to engage honestly with the struggle

Course Materials

Algebraic Topology

ISBN: 0-521-79540-0

Authors: Hatcher, Allen

Publisher: Cambridge University Press

Publication Date: 2002

Edition: online

Required/Recommended/Optional: Required (but free)

Additional Information

<https://pi.math.cornell.edu/~hatcher/AT/ATpage.html>

To enhance your learning and adequately prepare for all assignments and assessments, please consider all course materials listed above, along with additional content provided in class, posted to our course site, or included in the Course Schedule, as required reading and study unless indicated as optional.

Assessments and Assignments

There will be homework assignments approximately weekly/biweekly. You are encouraged to work together, but of course all submitted work must be your own. In addition, you either individually or in a small group will be responsible for a project. There will be both a short write up and a presentation component.

Activities and Grading

Activity	Quantity	%
Homework	5-6	60
Projects	1	40
Total		100%

Distribution of Course Assignments, Their Quantity, and Contribution to Final Grade.

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A+	
A	
A-	
B+	
B	
B-	
C+	
C	
C-	
D+	
D	
D-	
F	

Grade Distribution and Letter Grade

Course Guidelines for the Use of Generative Artificial Intelligence

AI tools such as large language models can be helpful for clarifying definitions, checking intuition, and exploring examples. You are permitted to use them in this spirit. However, submitting AI-generated proofs or solutions as your own work is academic dishonesty and defeats the purpose of the course. Algebraic topology is a subject where the struggle to write a proof is the learning; there are no shortcuts worth taking.

Course Expectations & Policies

Submission of late work must be communicated to me ahead of time. You may be asked to resubmit assignments to correct proofs etc.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

Department Policies & Resources

Ombudsperson

The ombudsperson is an advocate who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor.

Contact Ilse Rosales Martinez at ilse.rosalesmartinez@utsa.edu

Tutoring Services

UTSA provides many opportunities for student success, including tutoring services for specific math courses. The list below is subject to change based on availability.

MAT 1023, MAT 1043, MAT 1053, and MAT 1073

- We offer peer-to-peer academic support to students enrolled in a variety of courses. Both drop-in tutoring and one-on-one appointments are available.
- **Location:** [Tomás Rivera Center for Academic Excellence](#) MS 2.02.18 (behind the Rowdy statue)

MAT 1053 and MAT 1073

- Synergistic Targeted Academic Review Sessions (STARS), peer-led, interactive, and group focused sessions which cover traditionally difficult topics. The schedule will be provided by instructors through Canvas.
- **Location:** Determined by STARS tutors

MAT 1093, MAT 1213/1214, and MAT 1223/1224

- The Math Gym is unique in that it's a WeBWork lab designed to offer our STEM core students support from math majors and minors through targeted SI sessions. The schedule will be provided by instructors through Canvas
- **Location:** [Math Gym FLN 03.02.08](#)

Note: Course-specific support for upper-division courses may be available for a particular semester. Please request this information from your instructor.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.

- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit [Applying for Accommodations](#) or call (210) 458-4157.
- Your well-being is important. In addition to academic growth, life as a student can include many emotional experiences and opportunities. Being mindful of your own needs and available support can provide you with the tools to successfully navigate these experiences. Student support for well-being, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site for students. If needed, we can schedule a meeting, and I can connect you to resources. You are not alone; I am available to help you find resources that might best meet your needs.
- [Student Assistance Services \(SAS\)](#) is available to help you navigate academic and personal challenges. Campus Advocates in SAS are a central point of contact for students seeking clarity on university processes and policies. They also provide a direct connection to helpful resources and advocacy rooted in care. Campus Advocates offer confidential support and personalized guidance to help students succeed.
- The [Roadrunner Pantry](#) is here to help UT San Antonio students stay healthy and focused by providing free access to groceries, toiletries, and other essentials. With locations at the Main and Downtown campuses, the pantry is organized like a small store and offers items such as canned goods, snacks, and hygiene products. Students with a valid UTSA ID can visit the pantry to pick up what they need. Our goal is to ensure every Roadrunner feels supported and has the resources to succeed. If you're ever short on food or supplies, the Roadrunner Pantry is a welcoming space you can count on.
- Students at UT San Antonio are responsible for ensuring their work is consistent with UT San Antonio's standards for academic integrity. Students should review [Section 203 of the UT San Antonio Student Code of Conduct](#) for appropriate standards of academic integrity.
- UT San Antonio provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UT San Antonio Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open-enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UT San Antonio.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.